Although dynamic role assignments have been implemented in several systems, there is no clear standard for how to build dynamic roles. Approaches are usually based on database queries or perhaps based on rules specified in a logical form such as Prolog or the Web Ontology Language (OWL). When more flexible access control rules are required, attribute-based access control (ABAC) has largely replaced RBAC, as discussed in section 8.3. NIST have attempted to integrate ABAC with RBAC to gain the best of both worlds (https://csrc.nist.gov/publications/detail/journal-article/2010/adding-attributes-to-role-based-access-control), but this approach is not widely adopted.

Other RBAC systems implement constraints, such as making two roles mutually exclusive; a user can't have both roles at the same time. This can be useful for enforcing separation of duties, such as preventing a system administrator from also managing audit logs for a sensitive system.

EXERCISES

4. Which of the following are more likely to apply to roles than to groups?

a. Roles are usually bigger than groups.

b. Roles are usually smaller than groups.

c. All permissions are assigned using roles.

d. Roles better support separation of duties.

e. Roles are more likely to be application specific.